The IS-LM Model

Intermediate Macroeconomics

Fall 2025, Sciences Po. Paris Campus du Havre

Lectures 8

Announcements

- Midterm: Due on 23rd Oct, 2025

Introduction

- Previous lectures: goods market and financial markets analyzed respectively
- Today and next week: we look at goods and financial markets together, and analyze how output and the interest rate are determined in the short run
- John Hicks and Alvin Hansen called this framework the IS-LM model

The Goods Market and the IS Relation

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1. **Production / Output (Y):** $Y \uparrow \implies I \uparrow$ (see Solow model)
2. **Interest rates i:** Consider a firm wants to buy new machines $\implies$ it might need to borrow.
   If interest rates are too high, the firm might decide to borrow less $\implies I \downarrow$

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Graphically: ZZ line with 2 new features:

- Investment is no longer exogenous:
  - An increase in output leads to an increase in the demand for goods through its effects on <u>consumption</u> and <u>investment</u>
- ZZ is a curve (rather than a line)
  - because we do not assume that the consumption and investment relations are linear

The Derivation of the IS Curve

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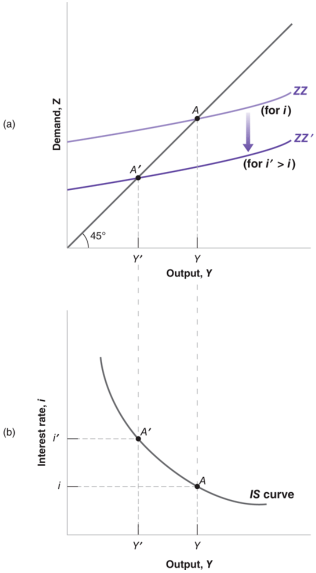

1. **An increase in the interest rate** $\Longrightarrow$ decreases the demand for goods at any level of output (*ZZ* shifts down) $\Longrightarrow$ a decrease in the equilibrium level of output.

2. Equilibrium in the goods market implies that an increase in the interest rate leads to a decrease in output.

The IS curve is therefore downward sloping

The IS curve

- The IS curve is so named because it documents the relationship between "investment" and "saving"
- The IS curve highlights the relationship between interest rates and output on the demand side of the economy (the goods market)
- Any point on the IS curve corresponds to the equilibrium in the goods markets

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Anything that causes C, I or G to change

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What shifts IS curve to the right?

(i.e., makes Y higher on the demand side of the economy)

- Increase in consumer confidence (stimulates C)
- Permanent increase in the wealth
- A reduction in taxes
- An increase in total factor productivity (stimulates I)
- An increase in government spending (i.e., war)
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Changes in interest rates **will not cause IS curve to shift**

$\Longrightarrow$ Changes in interest rates cause **movement along** IS curve!

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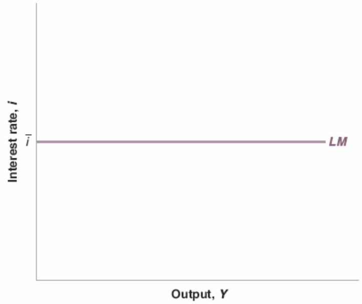
$\Longrightarrow$ **Liquidity-Money relation "LM relation"**

LM curve



The central bank chooses the interest rate (and adjusts the money supply so as to achieve it).

Any point on the LM curve corresponds to the equilibrium in financial markets ($M^s = M^d$)

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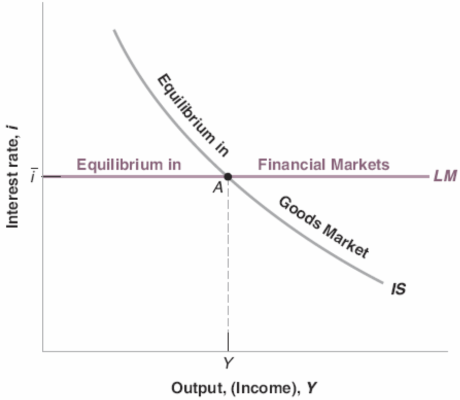
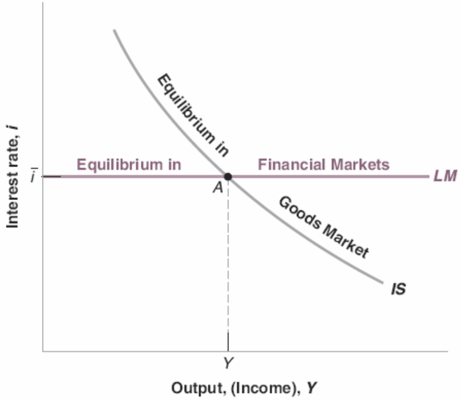
- Any point on the horizontal LM curve corresponds to equilibrium in financial markets

The IS-LM Model



- Equilibrium in the goods market is represented by the IS curve

The IS-LM Model



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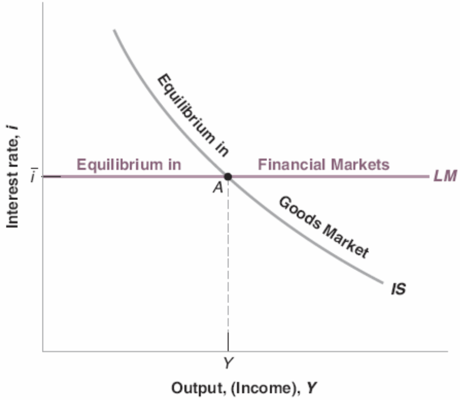

- Equilibrium in the goods market is represented by the IS curve
- Equilibrium in financial markets is represented by the LM curve
- Point A: both goods and financial markets in equilibrium

Why is this important?

The IS-LM model gives us a powerful framework to start thinking about the impact of:

- exogenous shocks (ΔC, ΔI ,ΔM)
- monetary policy
- fiscal policy

on the economy

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Steps for analyzing the effects of a changes in policy:

1. Does it shift the IS curve and/or the LM curve?
2. What does this do to equilibrium output and the equilibrium interest rate?
3. Describe the effects in words

Fiscal policy

Fiscal policy is related to **budget deficit**: $G - T$

- Increase in budget deficit [$\uparrow (G - T)$]: **Fiscal expansion**
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Let's take fiscal contraction:

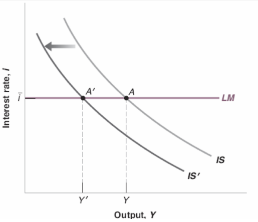
1. Increase in taxes
   or
2. Decrease in government spending

Fiscal policy

Fiscal contraction: Increase in taxes

$$IS: \; Y = C(Y - \color{red}{T}) + I(Y, i) + G$$
$$LM: \; i = \bar{i}$$

The Goods Market: $\uparrow$ taxes
$\implies$ lower disposable income
$\implies$ people decrease their consumption
$\implies$ lower demand
$\implies$ ($\times$ multiplier effect) lower output
IS: shift to the left
LM: movement along the curve



Monetary policy

Monetary policy is related to changes in interest rates

- Increase in interest rates [$\uparrow i$]: **Monetary contraction / Monetary tightening**
- Decrease in interest rates [$\downarrow i$]: **Monetary expansion**

Monetary policy

Monetary expansion: central bank decreases interest rates

$$IS: \; Y = C(Y - T) + I(Y, i) + G$$
$$LM: \; i = \bar{i}$$

Financial Markets:
$\downarrow$ interest rates
$\Longrightarrow$ LM: shift downwards

IS: movement along the curve
($\downarrow$ interest rates will increase investment
$\Longrightarrow$ $\uparrow$ output)

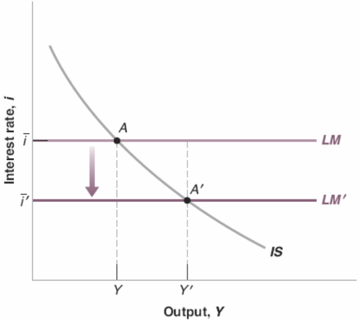


Figure: Caption

Monetary-Fiscal Policy Mix

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A combination of monetary and fiscal policies

1. MoPo and FiPo move the same direction
   - The economy is in a recession and output is too low
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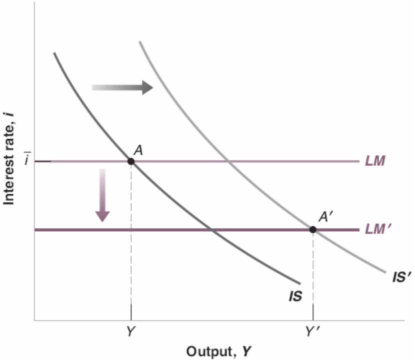
2. MoPo and FiPo move the opposite direction
   - For example: contractionary fiscal policy and expansionary monetary policy
   - When? Government wants to decrease its large budget deficit and the central bank intervenes to avoid the recession

Combined Fiscal and Monetary Expansion

- The fiscal expansion shifts the IS curve to the right
- The monetary expansion shifts the LM curve down
- Both lead to higher output
- Real-life example: The US recession of 2001



The U.S. Recession of 2001

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2. **Fiscal policy:**
   President George W. Bush cut taxes in 2001 and 2002 ($\downarrow T$)
   The events of 9/11 also lead to an increase in spending on defense and homeland security ($\uparrow G$)

A Short Note on the Transition Mechanism

Transition mechanism: a process by which general economic conditions are affected as a result of fiscal or monetary policy decisions

- What is affected? What is not?
- How strong are the effects? How strong are the side-effects?
- ...

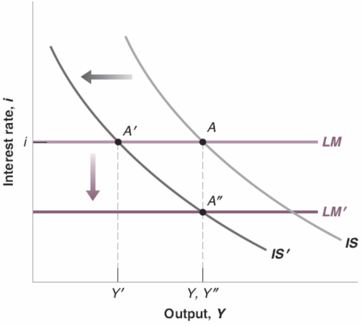
Fiscal and monetary policy have different effects on the composition of output

- $\downarrow T$: affect consumption (more than investment)
- $\downarrow i$: affect investment (more than consumption)

Opposite Direction of Economic Policies

Fiscal Consolidation and Monetary Expansion

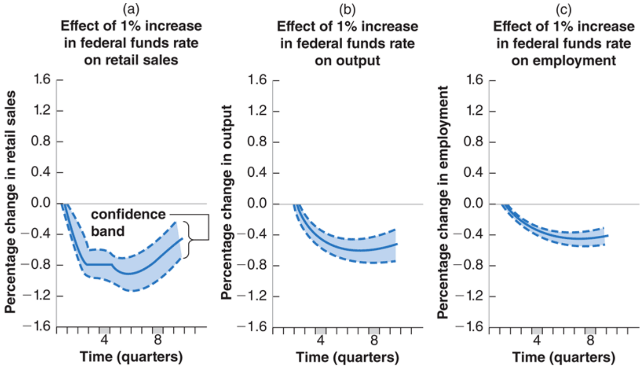
- The government wants to decrease the deficit:
  The fiscal consolidation shifts the IS curve to the left
- The Fed decides to intervene and prevent the economy from falling into recession:
  The monetary expansion shifts the LM curve down



The overall effect on output is not clear-cut
(in this graph: conveniently no effect, in reality: more complicated)

How Does the IS-LM Model Fit the Facts?

Monetary contraction (Cristiano et al., 1996)



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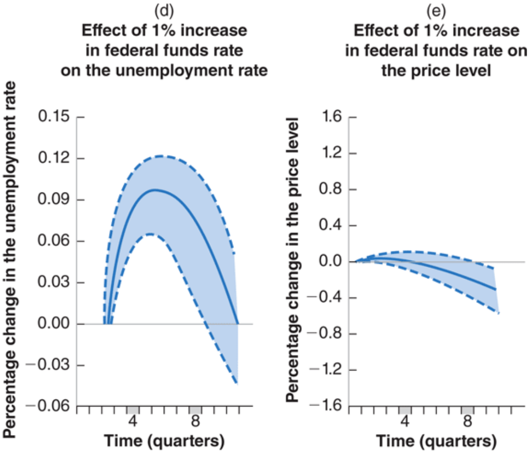

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